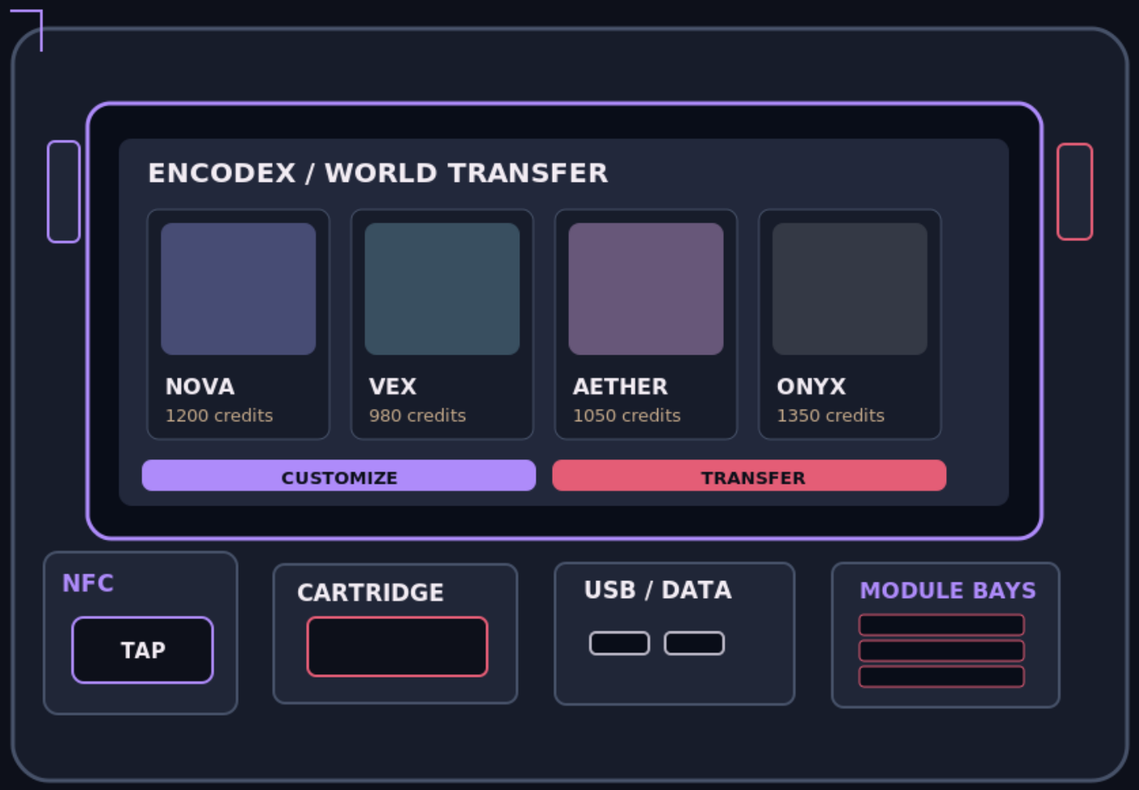
CHARACTER INTERSPHERE DOCK

V0.1 ENGINEERING CONCEPT / MODULAR NFC AVATAR TRANSFER TERMINAL

SHEET 01 / 02

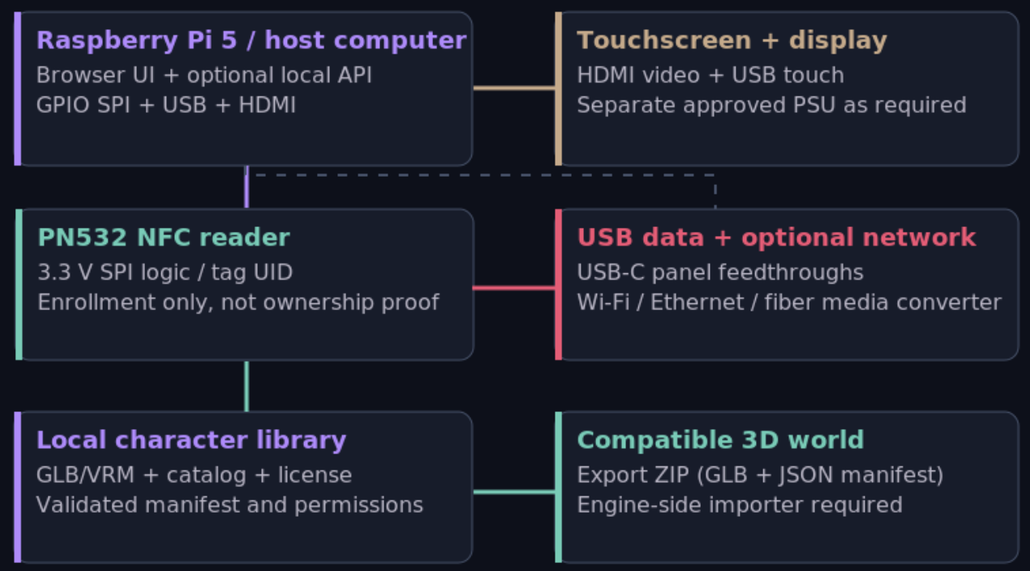
A / FRONT ELEVATION

Proposed envelope: 760 W x 420 D x 900 H mm



1. Flush-mounted touch display / HDMI + USB touch
2. Low-voltage NFC tap target / cartridge storage
3. Replaceable enclosure panels / non-load-bearing push rivets
4. Rear service door: cooling, power, network, cable strain relief

B / SYSTEM + DATA FLOW



C / PN532 SPI PIN MAP (PI 40-PIN HEADER)

Pi pin 1 : 3V3	———	PN532 : 3.3V
Pi pin 6 : GND	———	PN532 : GND
Pi pin 19 : GPIO10 / MOSI	———	PN532 : MOSI
Pi pin 21 : GPIO9 / MISO	———	PN532 : MISO
Pi pin 23 : GPIO11 / SCLK	———	PN532 : SCK
Pi pin 29 : GPIO5 / CE	———	PN532 : SSEL

Set PN532 SPI mode: SEL0 OFF / SEL1 ON. Enable SPI in Raspberry Pi OS.
GPIO is 3.3 V only. Do not connect powered 5 V or 12 V to GPIO.

D / POWER + SERVICE SAFETY

Separate certified 5 V / 5 A USB-C PSU for Pi 5; display uses maker-approved supply. No exposed mains inside kiosk.
Provide ventilation, mechanical standoffs, grommets, cable strain relief, an accessible shutdown, and service isolation.
Mechanical dimensions are concept targets; verify vendor drawings, accessibility, clearances, heat, and EMC before fabrication.

BILL OF MATERIALS / MODULAR ASSEMBLY

SHEET 2 · LOW-VOLTAGE REFERENCE BUILD / NOT A CERTIFIED WIRING DIAGRAM

01	**COMPUTE MODULE**	Raspberry Pi 5 (4GB or 8GB), cooled case/standoffs; OS + browser + Python server.
02	**DISPLAY / TOUCH**	HDMI touchscreen sized for front panel; USB touch data and approved external PSU.
03	**NFC INTERFACE**	Adafruit-compatible PN532 breakout set to SPI, with 3.3V logic and short shielded/routed wiring.
04	**PANEL I/O**	USB-C rated feedthrough(s), USB data hub as required, accessible Ethernet port.
05	**OPTICAL NETWORK**	Optional certified fiber-Ethernet converter; fiber carries network data, not device GPIO.
06	**POWER**	Certified Pi 5 USB-C 5V/5A PSU; separate display PSU if specified; no in-enclosure mains.
07	**MECHANICAL**	Powder-coated aluminum/steel sheet panels; snap-in non-structural trim; captive structural fasteners.
08	**THERMAL / SERVICE**	Forced-air ventilation with filter, service latch, strain relief and replaceable fan assembly.

NFC PHYSICAL PIN ROUTING

1 / 3V3 -> 3.3V; 6 / GND -> GND; 19 / GPIO10 -> MOSI; 21 / GPIO9 -> MISO; 23 / GPIO11 -> SCK; 29 / GPIO5 -> SSEL. Set PN532 SEL0 OFF, SEL1 ON for the Adafruit board.

FABRICATION NOTES

Proposed envelope: 760W x 420D x 900H mm. Verify actual display cutout, rear door swing, center of mass, tip resistance, accessibility, cooling and electromagnetic compliance.

CAUTION: Raspberry Pi GPIO is 3.3 V-only. Do not connect 5 V/12 V to GPIO, or use NFC UID as a purchase entitlement.

SOFTWARE & TRANSFER CONTROL

SHEET 3 · ENCODEX WORKFLOW / OFFLINE-FIRST REFERENCE ARCHITECTURE

01 **MARKETPLACE**

4 included demo 3D characters; local credit ledger; no payment processing.

v

02 **LIBRARY**

GLB/VRM byte storage; demo access flags; user imports stored in the session.

v

03 **ENCODEX TRANSLATOR**

UTF-8 / Base64 / Base64URL / Hex / Binary / URL / JSON string translation.

v

04 **TRANSFER PACKAGER**

ZIP: asset GLB or VRM + JSON manifest + import instructions.

v

05 **RECEIVER / WORLD**

Web / Unity / Unreal importer loads asset; target handles rig/controller and permissions.

IMPLEMENTED IN THE INCLUDED PROTOTYPE

Local UI, generated sample GLBs, browser-side ZIP, manifest export, local encoding, editable code templates, test-world scene and an optional loopback-only PN532 UID bridge.

NOT IMPLEMENTED / REQUIRES ENGINEERING

Real purchasing and entitlement service, authenticated remote 3D-world deployment, skinned humanoid animation, VRM-specific importer/rendering, hologram projection, and certified kiosk fabrication.